

Data Collection and Preprocessing Phase

Date	27 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	6 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains 1,048,575 transaction records with features related to online payments, used to predict fraudulent transactions. There were 1047433 fraudulent and 1142 non-fraudulent transactions.
Univariate Analysis	The univariate analysis of the dataset revealed key characteristics of individual variables. The isFraud column showed a highly imbalanced distribution, with 1,047,433 non-fraudulent transactions (99.89%) and only 1,142 fraudulent ones (0.11%). The feature, categorical with five unique values (e.g., CASH_OUT, TRANSFER), indicated that CASH_OUT and TRANSFER were the most common transaction types. Numerical features like amount exhibited a right-skewed distribution, with a mean around 180,000 and a lower median due to large transactions. Balance-related features (oldbalanceOrg, newbalanceOrig, oldbalanceDest, newbalanceDest) also showed skewness, with some transactions having zero or near-zero balances. The step feature, representing time steps, followed a sequential pattern without significant anomalies, suggesting a uniform temporal

	distribution.
Bivariate Analysis	Bivariate analysis explored relationships between pairs of variables to uncover patterns related to fraud. A strong correlation (~ 0.9) was observed between oldbalanceOrg and newbalanceOrig for non-fraudulent transactions, reflecting logical balance updates, whereas fraudulent transactions often showed anomalies, such as newbalanceOrig equaling zero despite large amount values. Fraudulent transactions were more prevalent in TRANSFER and CASH_OUT types compared to others, as seen in contingency tables or bar plots. The amount versus isFraud analysis indicated that fraudulent transactions typically involved higher amounts, evident in scatter plots. Numerical features like amount, oldbalanceOrg, and newbalanceDest showed moderate correlations (0.5–0.8), visualized through a heatmap, highlighting their interdependence in transaction patterns.
Multivariate Analysis	Multivariate analysis revealed complex patterns involving multiple variables. Fraudulent transactions were characterized by specific combinations, such as TRANSFER transactions with high amount and newbalanceOrig equaling zero, indicating potential fraud. Principal Component Analysis (PCA) and pair plots showed that non-fraudulent transactions formed dense clusters, while fraudulent ones appeared as outliers, suggesting distinct behavioral patterns. Feature interactions, such as the ratio of amount to oldbalanceOrg, were particularly indicative of fraud when unusually high, providing insights into predictive features. These patterns were visualized using scatter plot matrices.
Outliers and Anomalies	Outliers and anomalies were identified primarily in the amount and balance-related features. The amount feature contained extremely high values (e.g., above the 99th percentile, approximately 1M), detected through boxplots and Z-scores, which were retained due to their relevance to fraud detection. Anomalies in balance features, such as newbalanceOrig equaling zero despite large amounts of transfers, were flagged as potential fraud indicators, particularly in TRANSFER

	transactions. No outliers were removed, as they were critical for identifying fraudulent patterns, but robust scaling (e.g., using StandardScaler) was applied to mitigate their impact during modelling, ensuring the Random Forest Classifier could effectively handle these extreme values.
Data Preprocessing Code Screenshots	
Loading Data	<pre>import pandas as pd df = pd.read_csv('PS_20174392719_1491204439457_log.csv') print(df.head()) print(df.shape) print(df['isFraud'].value_counts())</pre>
Handling Missing Data	<pre>print(df.isnull().sum())</pre>
Data Transformation	<pre>from sklearn.preprocessing import LabelEncoder le = LabelEncoder() df['type'] = le.fit_transform(df['type']) df = df.drop(['nameOrig', 'nameDest'], axis=1)</pre>
Feature Engineering	<pre>fraud = df[df['isFraud'] == 1] non_fraud = df[df['isFraud'] == 0].sample(n=1288, random_state=42) df_balanced = pd.concat([fraud, non_fraud]) print(df_balanced['isFraud'].value_counts())</pre>
Save Processed Data	<pre>df_balanced.to_csv('balanced_dataset.csv', index=False)</pre>